

MATH 324 Winter 2025 - Tutorial 4

Section 003 (Thursday)

February 6, 2025

1 Consistency

Definition: $\hat{\theta}_n$ is a *consistent* estimator of θ if $\hat{\theta}_n \xrightarrow{p} \theta$. i.e.

$$\forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) = 0 \quad \text{or} \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| \leq \varepsilon) = 1.$$

Theorem 9.1 If $\hat{\theta}$ is unbiased for θ and $\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}_n) = 0$, then $\hat{\theta}_n \xrightarrow{p} \theta$.

Important properties for convergence in probability (i.e. $\hat{\theta}_n \xrightarrow{p} \theta$):

- Linearity: if $\hat{\theta}_n \xrightarrow{p} \theta$ and $\hat{\theta}'_n \xrightarrow{p} \theta'$, $a, b \in \mathbb{R}$, then $a\hat{\theta}_n + b\hat{\theta}'_n \xrightarrow{p} a\theta + b\theta'$.
- If $\hat{\theta}_n \xrightarrow{p} \theta$ and $\hat{\theta}'_n \xrightarrow{p} \theta'$, then $\hat{\theta}_n \hat{\theta}'_n \xrightarrow{p} \theta\theta'$. Similarly, if $\theta' \neq 0$, then $\hat{\theta}_n / \hat{\theta}'_n \xrightarrow{p} \theta / \theta'$.
- If the function g is continuous at θ , then $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$.

Question 1 - Law of Large Numbers

(Exercise 9.32) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from the probability density function

$$f(y) = \begin{cases} \frac{2}{y^2}, & \text{if } y \geq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Does the law of large numbers apply to $\bar{Y}_n$ in this case? Why or why not?

Solution

Weak law of large numbers states that $\bar{Y}_n \rightarrow \mu$ and $\text{Var}(\bar{Y}_n) \rightarrow 0$ as $n \rightarrow \infty$, under certain regularity conditions.

We assume that $\mathbb{E}[Y_i] = \mu < \infty$. Then, $\text{Var}(\bar{Y}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i)$ by independence.

$$\text{Var}(\bar{Y}_i) = \mathbb{E}[Y_i^2] - \mu^2 = \int_2^\infty y^2 \frac{2}{y^2} dy - \mu^2 = \int_2^\infty 2 dy - \mu^2$$

which is not finite. Hence the variance cannot go to 0 and we do not have the second result of WLLN.

We can notice that this distribution does not have a finite expectation, hence why WLLN cannot hold.

Question 2 - Consistency of sample mean with different variances

(Exercise 9.35) Let $Y_1, Y_2, \dots$ be a sequence of random variables with $\mathbb{E}[Y_i] = \mu$ and $\text{Var}(Y_i) = \sigma_i^2$. Notice that the σ_i^2 s are not all equal.

- (a) What is $\mathbb{E}[\bar{Y}_n]$?
- (b) What is $\text{Var}(\bar{Y}_n)$?
- (c) Under what condition (on the σ_i^2 s) can Theorem 9.1 be applied to show that $\bar{Y}_n$ is a consistent estimator for μ ?

Solution

- (a) $\mathbb{E}[\bar{Y}_n] = \mathbb{E}[\frac{1}{n} \sum_{i=1}^n Y_i] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] = \mu$.
- (b) $\text{Var}(\bar{Y}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2$
- (c) We have the first condition satisfied, which is that $\mathbb{E}[\bar{Y}_n] = \mu$. Then, $\text{Var}(\bar{Y}_n) \rightarrow 0$ if $\sum_{i=1}^n \sigma_i^2 < \infty$, i.e. all σ_i^2 are finite or $\max\{\sigma_1^2, \dots, \sigma_n^2\} < \infty$.

Question 3 - Consistency of continuous transformation

Show that if $\hat{\theta}$ is a consistent estimator of θ and h is a continuous function, then the sequence $h(\hat{\theta})$ is a consistent estimator of $h(\theta)$.

Solution

We'll denote $\hat{\theta}$ as $\hat{\theta}_n$ in the sequence $\hat{\theta}_1, \hat{\theta}_2, \dots$

By definition of continuity, h is continuous if $\forall \varepsilon > 0, \exists \delta > 0$ such that $|x - y| < \delta$, then $|h(x) - h(y)| < \varepsilon$ for any x, y in the domain.

Given $\varepsilon > 0$ (we are fixing ε needed for consistency), $\exists \delta > 0$ such that if $|\hat{\theta}_n - \theta| < \delta$, then $|h(\hat{\theta}_n) - h(\theta)| < \varepsilon$.

$\hat{\theta}_n \xrightarrow{P} \theta$, which means that $\lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| > \delta) = 0$, i.e. for a given $\alpha > 0, \exists N$ such that $\forall n \geq N$,

$$\mathbb{P}(|\hat{\theta}_n - \theta| > \delta) < \alpha.$$

We combine the two: $|\hat{\theta}_n - \theta| < \delta \Rightarrow |h(\hat{\theta}_n) - h(\theta)| < \varepsilon$ so

$$\mathbb{P}(|h(\hat{\theta}_n) - h(\theta)| > \varepsilon) \leq \mathbb{P}(|\hat{\theta}_n - \theta| > \delta) < \alpha$$

as long as $n \geq N$. This is true for any $\alpha > 0$, thus

$$\lim_{n \rightarrow \infty} \mathbb{P}(|h(\hat{\theta}_n) - h(\theta)| > \varepsilon) = 0.$$

2 Likelihood and Sufficiency

Definition: Suppose $Y_1, \dots, Y_n$ is a random sample from a distribution with parameter θ . The statistic $U = g(Y_1, \dots, Y_n)$ is *sufficient* for θ if $f(Y_1, \dots, Y_n|U)$, the conditional distribution of the Y_i 's given U , does not depend on θ .

Definition: Let $y_1, \dots, y_n$ be the observed values of the random sample $Y_1, \dots, Y_n$. The *likelihood* $L(y_1, \dots, y_n)$ is the joint probability/density of the discrete/continuous $Y_1, \dots, Y_n$ evaluated at $y_1, \dots, y_n$.

Theorem 9.4 (Factorization theorem) U is sufficient for θ if and only if

$$L(\theta) = L(y_1, \dots, y_n|\theta) = g(u, \theta)h(y_1, \dots, y_n).$$

That is to say, we can split the likelihood into: (1) a function that depends only on u and θ , and (2) a function without θ that may depend on the y_i .

Question 4 - Sufficiency of the power family

(Exercise 9.43) Let $Y_1, Y_2, \dots, Y_n$ denote independent and identically distributed random variables from a power family distribution with parameters α and θ . Then, if $\alpha, \theta > 0$,

$$f(y|\alpha, \theta) = \begin{cases} \alpha y^{\alpha-1}/\theta^\alpha, & 0 \leq y \leq \theta \\ 0, & \text{elsewhere} \end{cases}$$

If θ is known, show that $\prod_{i=1}^n Y_i$ is sufficient for α .

Solution

$$L(\alpha) = \prod_{i=1}^n f(y_i|\alpha, \theta) = \prod_{i=1}^n \frac{\alpha y_i^{\alpha-1}}{\theta^\alpha} = \frac{\alpha^n}{\theta^{n\alpha}} \prod_{i=1}^n y_i^{\alpha-1} = \frac{\alpha^n}{\theta^{n\alpha}} \left(\prod_{i=1}^n y_i \right)^{\alpha-1}$$

Therefore, $g(u, \alpha) = \frac{\alpha^n}{\theta^{n\alpha}} u^{\alpha-1}$ and $h(y_1, \dots, y_n) = 1$. By theorem 9.4, $U = \prod_{i=1}^n y_i$ is sufficient for α .

Question 5 - Sufficiency of the exponential

(Exercise 9.46) If $Y_1, Y_2, \dots, Y_n$ denote a random sample from an exponential distribution with mean β , show that $f(y|\beta)$ is in the exponential family and that Y is sufficient for β .

Solution

The exponential family has the form

$$f(y|\theta) = \begin{cases} a(\theta)b(y)e^{-[c(\theta)d(y)]}, & \alpha \leq y \leq \beta \\ 0, & \text{elsewhere} \end{cases}$$

where α, β may be $\pm\infty$.

For $Y \sim \text{Exp}(\beta)$,

$$f(y|\beta) = \begin{cases} \frac{1}{\beta} e^{-y/\beta}, & y \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

Question 6 - Sufficiency of the uniform

(Exercise 9.49) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from the uniform distribution over the interval $(0, \theta)$. Show that $Y(n) = \max(Y_1, Y_2, \dots, Y_n)$ is sufficient for θ .

Solution

For the uniform,

$$f(y|\theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq y \leq \theta \\ 0, & \text{elsewhere} \end{cases}$$

Let

$$I_{(a,b)}(y) = \begin{cases} 1, & \text{if } a \leq y \leq b \\ 0, & \text{otherwise.} \end{cases}$$

This is the indicator function for the interval (a, b) . Then, we can write

$$f(y|\theta) = \frac{1}{\theta} I_{(0,\theta)}(y).$$

Our likelihood is

$$L(\theta) = \prod_{i=1}^n f(y_i|\theta) = \prod_{i=1}^n \frac{1}{\theta} I_{(0,\theta)}(y_i) = \frac{1}{\theta^n} \prod_{i=1}^n I_{(0,\theta)}(y_i)$$

$\prod_{i=1}^n I_{(0,\theta)}(y_i) = 0$ if any of the y_i 's are outside of $[0, \theta]$, i.e. if any of the $y_i > \theta$. Thus if $y_{(n)} > \theta$, then the product is 0; if $y_{(n)} \leq \theta$, then the rest are also less than θ and the product is 1. Hence,

$$L(\theta) = \frac{1}{\theta^n} I_{(0,\theta)}(y_{(n)})$$

which means $g(u, \theta) = \frac{1}{\theta^n} I_{(0,\theta)}(u)$ and $h(y_1, \dots, y_n) = 1$. By the factorization theorem, $Y_{(n)}$ is sufficient for θ .